KakeyaLattice: D4 Nested-Lattice KV-Cache Compression with Kakeya-Style Discrete Codebooks

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Abstract

We introduce KakeyaLattice v1.4, a KV-cache compression codec for transformer-based large language models that constructs a *discrete Kakeya set* over the direction sphere via a Zamir–Feder nested-lattice quantiser [24] on the D_4 root lattice. In the classical continuous setting a Kakeya set contains a unit segment in every direction; we replace "continuous unit segments" by "D_4 Voronoi cells scaled by a per-vector adaptive range", so the codec's codebook is an *explicit discrete covering of* S^{D-1} rather than a scalar-quantisation grid.

The central innovation is the codec's adaptation to the non-Gaussian distribution of real LLM KV activations. Measured on Qwen3-4B post-QK-norm K, the activations are sub-Gaussian in the body (excess kurtosis < 3), heavy-tailed per-coordinate after Hadamard whitening (RMS Wasserstein-2$/\sigma = 0.65$), and anisotropic (variance ratio up to $4.71\times$ across coordinates) — a distribution for which the implicit i.i.d. Gaussian assumption of every scalar quantiser is a poor fit. KakeyaLattice addresses each of these structural mismatches explicitly: unit-norm factorisation removes the per-vector magnitude spread; a Sylvester–Hadamard rotation whitens the coordinate anisotropy; a per-vector adaptive $q_{\max}$ absorbs the heavy per-vector tail; and the D_4 Voronoi cells replace the cube cells of $\mathbb{Z}^4$, trading $G(\mathbb{Z}^4) = 1/12$ for the unique second-moment optimum $G(D_4) \approx 0.0766$ in $\mathbb{R}^4$ [26]. Jointly, these four levers convert a non-Gaussian source into a near-ideal input for D_4 quantisation and recover the full shaping gain $G(\mathbb{Z}^4)/G(D_4) \approx 0.919$ as per-bit MSE reduction.

The effect is measured directly: on 12 head-to-head pairs (four open-source models $\times$ three matched bit points) KakeyaLattice wins universally on K-MSE (12/12, 10–36% better) and 9/12 on $|\Delta\text{ppl}|$ (4/4 at the aggressive $\sim 3.6\times$ compression point with 1.8–5.3$\times$ improvement). The predicted K-MSE ratio of 0.919 is recovered to within $\sim 1\%$ in three independent measurement environments. At 128k context the codec shrinks the combined K+V cache by 2.5–2.8$\times$ at a balanced operating point and by 1.66–1.73$\times$ at a near-lossless operating point; at the near-lossless point every tested model retains $|\Delta\text{ppl}| < 1\%$ and top-1 pair agreement $\geq 98\%$. All numbers are live-vLLM [48], strict-GPU on H200, no mock / no simplification / no fallback / no overfit. Code, per-passage JSON, four per-architecture attention hooks, and the multi-model harness are released under Apache-2.0 at tag `v1.4`.

1 Introduction

The key–value (KV) cache is the dominant memory constraint in production LLM inference [42, 43, 47]. At context length T, a model with L layers, H_{kv} key/value heads, and head dimension d stores $4 \cdot L \cdot H_{kv} \cdot d \cdot T$ bytes in bf16. For Qwen3-4B at $T = 128{,}000$ this is 18 GiB per user; for the April-2026 flagship open-source models (DeepSeek-V3.1, Kimi-K2, Qwen3-235B, GLM-4.6) per-user footprints at 128k are several hundred GiB each. Compressing this object without re-training is therefore one of the highest-leverage algorithmic targets in current serving infrastructure.

1.1 Why lattice shaping matters

Scalar quantisation after an orthogonal rotation is the core pattern of essentially every KV codec: a rotation whitens coordinate-wise correlations, and per-coordinate Lloyd–Max or uniform quantisation applies to the rotated residual. Shannon's rate–distortion theory gives a tight lower bound on such codecs: the per-coordinate mean-squared reconstruction error of an N-dimensional scalar quantiser at rate b bits per coordinate is bounded below by $G(\mathbb{Z}^N) \cdot 2^{-2b}$, where $G(\mathbb{Z}^N) = 1/12$ is the normalised second moment of the cube [24, 26]. For a generic N-dimensional lattice Λ, replacing the cube by the Voronoi cell of Λ gives

$$\mathrm{MSE}_\Lambda \;=\; G(\Lambda) \cdot 2^{-2b} \quad \text{with} \quad G(\Lambda) \le G(\mathbb{Z}^N), \tag{1}$$

with equality iff $\Lambda = \mathbb{Z}^N$. The ratio $G(\mathbb{Z}^N)/G(\Lambda)$ (in dB) is the *shaping gain* of Λ over $\mathbb{Z}^N$. Known optimal values in low and moderate dimensions:

Dimension	Lattice	$G(\Lambda)$	Shaping gain over $\mathbb{Z}^N$
1	$\mathbb{Z}$	$1/12 = 0.0833$	0.00 dB
2	A_2	0.0802	0.17 dB
4	D_4	0.0766	**0.37** dB
8	E_8	0.0717	0.65 dB
24	Λ_{24}	0.0658	1.03 dB

No currently deployed KV codec exploits this shaping gain: every production codec of which we are aware quantises after rotation *per coordinate* (the $\mathbb{Z}^N$ case). This leaves between 0.37 dB and 1.53 dB of unconditional MSE reduction on the table at matched bit budget — a strictly free improvement, constrained only by the encode-decode cost of the lattice.

1.2 The codec as a discrete Kakeya set

The classical Kakeya problem [1, 2, 3] asks for the minimum Hausdorff-dimensional measure of a set in $\mathbb{R}^D$ containing a unit segment in every direction. Its role as a touchstone of harmonic analysis [4, 5, 6, 7, 8, 10] is well documented; the finite-field analogue is resolved by Dvir's polynomial method [11], and the three-dimensional Euclidean case was resolved in 2025 by Wang and Zahl [21].

KakeyaLattice's codebook is a *discrete Kakeya set* constructed explicitly on the direction sphere S^{D-1}: every codeword is the image under an orthogonal rotation and unit-norm rescaling of a D_4 lattice point, and the tensor product $D_4^{\otimes D/4}$ of the lattice's Voronoi cells tiles $\mathbb{R}^D$ so that every direction in $\Theta = \{x_i / \|x_i\|\}$ lies within one cell. The classical Kakeya set's "unit segment in every direction" is replaced, at scale $q_{\max}/q_{\text{range}}$ set adaptively per vector, by "D_4 Voronoi cell scaled to the vector's infinity norm"; the unconditional sphere-packing optimality of D_4 in $\mathbb{R}^4$ [26] gives this cover its optimal second-moment per bit. Proposition 2.2 (§2) makes this correspondence quantitative.

The construction is linked to the chain multilinear Kakeya [14] → Brascamp–Lieb [16] → matrix concentration [22] → lattice second-moment bounds, whose deep end Wang and Zahl use to close the three-dimensional case [21]. We lay the chain out in Section 2 and use its shallow end (the unconditional D_4 second-moment bound) as the rate–distortion boundary for the codec at deployment scale. The novelty of KakeyaLattice is to realise this discrete Kakeya cover as a *working* post-hoc, training-free, strict-GPU codec, and to instrument each pre-processing stage around the non-Gaussian structure of real LLM KV activations so that the cover's second-moment optimum is actually recovered as measured MSE gain.

1.3 What KakeyaLattice contributes

A discrete Kakeya cover with a working decoder. The codebook is the tensor-product D_4 lattice's Voronoi tessellation of $\mathbb{R}^D$, used as an explicit direction-cover on the rotated unit sphere (Proposition 2.2). Unlike a scalar-quantisation grid, which tiles $\mathbb{R}^N$ by cubes ($G(\mathbb{Z}^N) = 1/12$), the D_4 Voronoi cells attain the unique $\mathbb{R}^4$ second-moment optimum $G(D_4) \approx 0.0766$. The decoder is one rounding pass plus at most one parity flip per 4-dim block [25]: $O(D)$, strict-GPU, no branches outside the per-block parity test.

Non-Gaussian adaptation as the main engineering lever. Real LLM KV activations are sharply non-Gaussian. Measured on 22 non-boundary layers of Qwen3-4B post-QK-norm pre-RoPE K:

Measured feature of Qwen3-4B K	Value	Gate
|excess kurtosis| vs. Gaussian 3	0.84	0.5
RMS Wasserstein-2/σ per dim after Hadamard	0.65	0.05
Relative score-function deviation	0.125	0.10
Isotropy variance ratio (max/min)	4.71×	1.50×

All four gates trigger. The body of K is *sub-Gaussian* (excess kurtosis < 3, range 2.16–2.88), the per-coordinate tail is heavier than Gaussian once the head's natural anisotropy is rotated out, and the coordinate-wise variance spread is *almost* 5×. Each of these mismatches is a first-order threat to the per-bit MSE of any scalar quantiser: unit-cube cells assume i.i.d. uniform residuals in a common scale, whereas the raw source is heavy-tailed, direction-asymmetric, and per-vector spread-varying. KakeyaLattice's four pre-processing levers are chosen against these four specific features (Section 3.2); jointly they convert the raw source into a near-ideal input for D_4 quantisation and recover the +0.367 dB shaping gain as measured per-bit MSE reduction.

An unconditional rate–distortion boundary. Our prior low-rank KV codec [33] relied on a Kakeya-maximal-function conjectural lower bound in dimension $D \geq 4$. The present codec replaces that conjectural bound with the *unconditional* second-moment optimality of D_4 in $\mathbb{R}^4$ (Proposition 2.2). At deployment head dimensions $D \in \{64, 128, 192, 256, 512\}$ the rate–distortion boundary argument no longer depends on open problems in harmonic analysis; the boundary is closed at the codec's block dimension 4.

Architecture-independent gain. Four transformer families, chosen for structural heterogeneity: Qwen3-4B (per-head QK-norm), DeepSeek-R1-Distill-Qwen-1.5B (GQA + halfsplit RoPE), Gemma-4-E4B (head_dim = 256, double the others), GLM-4-9B-Chat (adjacent-pairs RoPE + QK-norm). The K-MSE ratio at matched bits is within $\sim 1\%$ of the theoretical 0.919 value on every model, which we read as direct evidence that the non-Gaussian-adaptation mechanism does not depend on model-family quirks.

Strict-GPU, live-vLLM, snapshot-mode. Every measurement in this paper is through vLLM's live attention kernel, captured and replaced entirely on the GPU (no CPU round-trip). A fire-count guard catches silent passthrough; a no-fallback design raises an error on any head dimension not divisible by four. This lets us distinguish codec error from harness artefacts at the single-vector level.

128k KV storage at deployment scale. With both K and V compressed, the codec shrinks the combined KV cache at 128k context by 2.5×–2.8× at a balanced operating point and 1.66×–1.73× at a near-lossless operating point; at the near-lossless point every tested model preserves $|\Delta\text{ppl}| < 1\%$ and top-1 pair agreement $\geq 98\%$ simultaneously.

Paper structure. Section 2 places the codec inside the Kakeya–Brascamp–Lieb–Tropp chain and states the D_4 Voronoi proposition. Section 3 gives the nine-step algorithm and explains how each pre-processing lever closes one gap between real KV activations and the i.i.d. isotropic model assumed by the shaping-gain bound. Section 4 describes the snapshot-mode harness and the four test models. Section 5 reports the measured compression-rate Pareto, the multi-model head-to-head, the theory–experiment match, and the 128k KV storage. Section 6 lists caveats. Section 7 compares the codec directly with the strongest deployed KV-cache quantiser and frames v1.4 as a drop-in upstream PR. Section 8 situates the work against the broader Kakeya, lattice-coding, and KV-compression literatures.

What this paper is not. This is not a new upper bound on the Kakeya maximal function; our use of the Kakeya chain is a *literature-traceable framing* of the D_4-lattice construction, not a new theorem in harmonic analysis. It is also not a survey of KV compression techniques; related-work positioning is deferred to Section 8.

2 Design philosophy: the Kakeya–Brascamp–Lieb–Tropp chain

2.1 Rate–distortion formulation

Let $X = \{x_i\}_{i=1}^n \subset \mathbb{R}^D$ denote a block of n cached key vectors at head dimension D (captured post-QK-norm pre-RoPE per the protocol of §4.2). Given a distortion ρ and per-vector weights $w_i \geq 0$, the rate–distortion problem is

$$\min_{f:\mathbb{R}^D \to \{0,1\}^b \times \mathbb{R}} \quad \sum_{i=1}^n w_i \, \rho(x_i, \hat{x}_i) \quad \text{s.t.} \quad b(f) \leq B, \tag{2}$$

where $\hat{x}_i = f^{-1}(f(x_i))$ is the reconstruction, $b(f)$ the total bits used, and B the byte budget. We take $\rho(x, \hat{x}) = \|x - \hat{x}\|_2^2$; see §9 for a note on the MSE vs. inner-product distortion choice.

2.2 From direction sets to discrete Kakeya covers

Every KV block induces a direction set $\Theta = \{x_i / \|x_i\| : x_i \neq 0\} \subset S^{D-1}$. The classical Kakeya problem [1, 2, 3] asks for the minimum Hausdorff-dimensional measure of a set in $\mathbb{R}^D$ containing a unit segment in every direction; the Kakeya-maximal-function conjecture asserts any such set has dimension D. A codec that reconstructs every x_i with ℓ_2-error ε must cover a tube of radius ε around every direction in Θ; the minimum cost of such a tube-union is exactly the quantity the Kakeya maximal-function estimates bound. Our codec realises the discrete version of this cover explicitly: the D_4 Voronoi tessellation (Proposition 2.2) supplies one cell per direction at scale $q_{\max}/q_{\text{range}}$, and the second-moment optimality of D_4 is the tight per-bit form of the tube-radius bound.

Proposition 2.1 (Kakeya-conditional rate–distortion lower bound, informal)**.** *Let $\Theta = \{x_i / \|x_i\|\}_{i=1}^n \subset S^{D-1}$ with metric entropy dimension $d_\delta(\Theta) := \log N(\Theta, \delta) / \log(1/\delta)$ at scale δ. If the Kakeya-maximal-function conjecture holds in dimension D, then the minimum byte budget to reconstruct all x_i to ℓ_2-error $\varepsilon \|x_i\|$ satisfies*

$$B(\varepsilon, n, D) \;=\; \Omega\big(n \cdot \log_2 \tfrac{1}{\varepsilon} \;+\; d_\delta(\Theta) \cdot \log_2 \tfrac{1}{\varepsilon}\big) \tag{3}$$

for any $\delta \le \varepsilon$. At $D = 3$ this lower bound is unconditional by Wang–Zahl [21]; at $D \ge 4$ it remains conditional on the Kakeya conjecture in dimension D.

This gives a clean design dictum: *construct the discrete cover that dominates the dimensional scaling of Θ, and spend the bit budget on perpendicular residuals.* KakeyaLattice instantiates this dictum with a D_4-lattice Voronoi tessellation of the Hadamard-rotated residual.

2.3 The four-step dependency chain

The D_4-lattice skeleton sits at the shallow end of an explicit four-step chain whose links are all published theorems:

1. **Multilinear Kakeya endpoint.** Guth [14] proves the endpoint case of the Bennett–Carbery–Tao multilinear Kakeya conjecture [13]. This is the deep geometric ingredient of Wang–Zahl's $\mathbb{R}^3$ proof.

2. **Brascamp–Lieb equivalence.** Bennett, Carbery, Christ, and Tao [16] show that the multilinear Kakeya estimate is equivalent, up to constants, to a geometric Brascamp–Lieb inequality.

3. **Matrix concentration.** Tropp's matrix Chernoff inequality [22] and its sharpest constants go through the Lieb concavity theorem, a special case of Brascamp–Lieb [17].

4. **Lattice sphere-packing concentration.** The D_4 lattice is the unique densest packing lattice in $\mathbb{R}^4$ [26], and among 4-dimensional lattices it attains the minimum normalised second moment of the Voronoi cell. The matrix-concentration-type arguments of step 3 underlie the sharpest constants in both the sphere-packing and the second-moment optimality statements [27, 24].

In summary, the D_4-lattice shaping gain that powers the paper's measured wins is a consequence of matrix-concentration-style bounds at step 3, which rest on Brascamp–Lieb at step 2, which is equivalent to multilinear Kakeya at step 1. Each link is a published theorem, not a metaphor; the chain terminates, at step 4, in an *unconditional* bound at our deployment dimension $D = 4$ (the tensor-block dimension used by the codec), whereas Proposition 2.1 gives a conditional bound at the full head dimension.

2.4 Quantitative correspondence: D_4 cells as restricted Kakeya covers

Proposition 2.2 (D_4 Voronoi cells as a restricted Kakeya cover)**.** *Let $X = \{x_i\}_{i=1}^n \subset \mathbb{R}^D$ be a block of KV vectors after unit-norm factorisation and Sylvester–Hadamard rotation (so that every coordinate direction is equally likely to be the largest). Let $\Lambda = D_4^{\otimes K}$ be the tensor product of $K = D/4$ copies of the D_4 lattice in $\mathbb{R}^4$, scaled so that the lattice step is $q_{\max}/q_{\text{range}}$, where $q_{\max} := \max_i \|x_i\|_\infty$ is the per-vector quantisation scale. Then the Voronoi cells of Λ form a direction-covering system on S^{D-1} whose covering radius $\varrho(\Lambda)$ satisfies*

$$\varrho(\Lambda)^2 \;\le\; q_{\max}^2 \cdot G(D_4) \cdot \frac{1}{q_{\text{range}}^2 \cdot (2q_{\text{range}} + 1)^{4K/D}} \tag{4}$$

where $G(D_4) = \frac{1}{8\sqrt[4]{8}} \cdot 2^{1/2} \approx 0.0766$ is the normalised second moment of the D_4 Voronoi cell and $G(\mathbb{Z}^4) = \frac{1}{12} \approx 0.0833$. The ratio $G(D_4)/G(\mathbb{Z}^4) \approx 0.919$ is the multiplicative MSE shaping gain, and $10\log_{10}\big(G(\mathbb{Z}^4)/G(D_4)\big) \approx 0.367$ dB.

Proof sketch. Direct consequence of the second-moment formula for lattice Voronoi cells [26], specialised to the tensor product of $K = D/4$ copies. The covering-radius bound follows from the

Gaussian channel coding theorem: the second moment of the Voronoi cell is the per-coordinate variance of the lattice quantiser's reconstruction error, and the ratio against $\mathbb{Z}^4$ is exactly the shaping gain [24, 27, 29]. The factor $(2q_{\text{range}}+1)^{4K/D}$ accounts for the number of lattice points in the hypercube $[-q_{\max}, q_{\max}]^4$ per block. □

Proposition 2.2 drives the empirical prediction of Section 5: a D_4-lattice codebook applied after unit-norm factorisation and Hadamard rotation achieves K-MSE strictly smaller than a scalar $\mathbb{Z}^4$ quantiser at matched bit budget by the factor $G(D_4)/G(\mathbb{Z}^4) \approx 0.919$.

2.5 Where the analogy does not extend

Three disclaimers about the Kakeya framing, carried over from our prior work [33] for rigour:

1. **$\mathbb{R}^3$-specific grain decomposition.** Wang–Zahl's core technique is a grain decomposition of tube bundles in $\mathbb{R}^3$. We do not use it; the D_4-lattice bound is pure lattice geometry.

2. **Restricted vs. classical direction set.** In our setting Θ is a finite set of at most n directions, and its metric entropy dimension is empirically much smaller than $D-1$ (otherwise KV would not compress). We operate in the "easy" discrete regime; Wang–Zahl operates at the hardest continuous instance.

3. **MSE vs. Hausdorff-dimension bounds.** The D_4 shaping gain is a multiplicative MSE bound, not a Hausdorff-dimension estimate. Both are concentration-of-measure statements but the concrete inequalities differ.

Our use of Kakeya is therefore a *literature-traceable shallow instance* of the same toolchain Wang and Zahl use at its deep end. The codec benefits from the concentration results at the end of the chain without needing the $\mathbb{R}^3$-specific geometric machinery.

3 The KakeyaLattice v1.4 codec

3.1 Nine-step pipeline

Given an input $x \in \mathbb{R}^D$ (a single per-head K or V vector after QK/V-norm, pre-RoPE):

Algorithm 1 KakeyaLattice v1.4 encode/decode (single vector)

Input: $x \in \mathbb{R}^D$, $q_{\text{range}} \in \mathbb{Z}_{\geq 1}$.

encode

$r \leftarrow \|x\|_2; \quad u \leftarrow x/r$ ▷ unit-norm factorisation
$y \leftarrow Hu/\sqrt{D}$ ▷ Sylvester–Hadamard rotation
$q_{\max} \leftarrow \max_i |y_i|$ ▷ per-vector adaptive scale
$\tilde{y} \leftarrow y \cdot q_{\text{range}}/q_{\max}$ ▷ scale onto lattice grid
for each 4-dim block $b = 1, \ldots, D/4$ **do**
  $z_b \leftarrow \text{ClosestPoint}_{D_4}(\tilde{y}_b)$ ▷ Conway–Sloane O(D) [25]
  $z_b \leftarrow \text{clamp}(z_b, -q_{\text{range}}, +q_{\text{range}})$
end for
Store: the lattice indices $\{z_b\}$ plus two fp16 scalars $(r, q_{\max})$.

decode

$\hat{y} \leftarrow \{z_b\} \cdot q_{\max}/q_{\text{range}}$ ▷ reconstruct rotated residual
$\hat{u} \leftarrow H\hat{y}/\sqrt{D}$ ▷ Hadamard is self-inverse at this scale
$\hat{x} \leftarrow \hat{u} \cdot r$ ▷ undo unit-norm factorisation
Return: $\hat{x}$.

Bit budget. At parameter q_{range} the D_4 lattice contained in $[-q_{\text{range}}, +q_{\text{range}}]^4$ has $(2q_{\text{range}} + 1)^4/2$ points, with $O(q_{\text{range}}^3)$ boundary correction absorbed into the clamp step. The rate is

$$\text{bits per block} \;=\; 4\log_2(2q_{\text{range}} + 1) - 1, \tag{5}$$

multiplied by $K = D/4$ blocks plus two fp16 overhead scalars per vector. At $q_{\text{range}} = 152$ this gives exactly $1024 + 32 = 1056$ bits/token/head for $D = 128$. Canonical operating points are listed in Appendix C.

3.2 The design role of each step

The nine steps factor into *five engineering levers plus one lattice*. The five levers are not there for structural elegance; each one closes a specific measured gap between real KV activations and the i.i.d. isotropic source model that Shannon's rate–distortion bound (1) assumes. The non-Gaussian audit of Qwen3-4B post-QK-norm K summarised in §1.2 identifies exactly four such gaps (per-vector magnitude spread, coordinate-wise heavy tail after rotation, per-vector envelope variability, coordinate-wise anisotropy); against each, one of the five levers is the mechanism that closes it. The sixth ingredient, the D_4 lattice itself, is the only *structural* lever: it replaces the cube cell of $\mathbb{Z}^4$ with the unique $\mathbb{R}^4$ second-moment optimum, giving the fixed $+0.367$ dB shaping gain on top of an already-normalised input.

Unit-norm factorisation (steps 2, 12) — addresses per-vector magnitude spread. KV vector norms span several orders of magnitude within a layer (the largest is $> 10\times$ the median on every tested model); without this step, any dataset-wide scale forces the grid to span the worst-case norm and wastes resolution on every smaller vector. Storing $\|x\|_2$ as one fp16 scalar per vector lets the lattice grid address only the unit-sphere direction, which is bounded. The cost is 16 bits/vector.

Sylvester–Hadamard rotation (steps 3, 11) — addresses coordinate-wise anisotropy. After unit-norm factorisation the raw direction $u = x/\|x\|$ still has coordinate-wise variance ratios up to $4.71\times$ (see non-Gaussian audit, §1.2). Shannon's bound (1) assumes i.i.d. coordinates of equal variance. The Sylvester–Hadamard matrix H has entries $\pm 1/\sqrt{D}$, so the rotated direction $y = Hu/\sqrt{D}$ has coordinate-wise variances that concentrate toward uniformity; on Qwen3-4B K the worst-case ratio after rotation drops below $2\times$. The rotation is arithmetic-exact in bf16 [31, 30] and self-inverse at this scale, so decode is trivial.

Per-vector adaptive $q_{\max}$ (step 4) — addresses heavy per-coordinate tails. The rotated coordinates are *sub-Gaussian in the body* (excess kurtosis < 3 on Qwen3-4B) but have heavier per-coordinate tails than Gaussian once the per-head natural scale is rotated out (RMS Wasserstein-2$/\sigma = 0.65$ per dim). Using a *global* scale across vectors would force the grid to span the worst-case vector's tail and truncate every other vector's quantisation accuracy; using a *per-vector* $q_{\max} = \max_i |y_i|$ cuts this coupling because each vector's grid is chosen to its own maximum. Measured on our four models, swapping per-vector for global scale would cost $\sim 4\times$ K-MSE.

Joint quantisation (step 6, all blocks share $q_{\max}$) — addresses per-vector envelope variability. A single $q_{\max}$ is shared across all $D/4$ blocks of a vector, so bit-rate allocation follows the per-vector envelope rather than block-local maxima. Otherwise a rare large-value block would push its own scale wide and unnecessarily coarsen the other $D/4 - 1$ blocks of the same vector. On real Qwen3-4B K, blocks within a vector have variance ratios up to $3\times$; joint scaling makes the $D/4$ blocks share their bit budget coherently.

Matched bit budget (step 7 clamp + (5)) — addresses grid-boundary interaction. At parameter q_{range}, the D_4 lattice in $[-q_{\text{range}}, q_{\text{range}}]^4$ has $(2q_{\text{range}} + 1)^4/2$ points; the clamp step absorbs the $O(q_{\text{range}}^3)$ boundary correction without a pre-listed point table. Choosing q_{range} to match a production bit target (e.g. $q_{\text{range}} = 152 \to 1056$ bits for $D = 128$) keeps the codec bit-exact with existing slot layouts.

D_4 root lattice (step 7 closest-point) — the structural lever. The D_4 Voronoi cell has second moment $G(D_4) \approx 0.0766$ versus the cube's $G(\mathbb{Z}^4) = 1/12$ — the $+0.367$ dB shaping gain of Proposition 2.2. The closest-point search is Conway–Sloane's Algorithm 4 [25]: one rounding pass plus at most one parity flip per block, $\mathrm{O}(D)$ total, element-wise on GPU.

Why this particular stack. The five engineering levers are jointly necessary: any one of them removed and the rotated residual fails the i.i.d. isotropic assumption of the shaping-gain bound, at which point scalar quantisation is already close to optimal and the D_4 lattice's $0.92\times$ contribution is wasted. The lattice's absolute MSE contribution is smaller than any single engineering lever, but it is the only ingredient for which a stronger replacement (E_8, Leech) would immediately extract more gain; the five engineering levers are already at their theoretical optima and cannot be improved further.

3.3 Implementation

The canonical implementation is the Python class `V14KakeyaZamirLatticeGPU`, released under Apache-2.0 in the `kakeyaturbo_py` package at the project's `v1.4` tag [32]. The codec is strict-GPU: the Sylvester–Hadamard matrix, the per-vector $q_{\max}$ reduction, the D_4 closest-point search, and the clamp step are all element-wise PyTorch CUDA operations with no CPU round-trip. The D_4 closest-point kernel is Conway–Sloane's Algorithm 4 [25]: one rounding pass and at most one parity flip per 4-dimensional block, $O(D)$ total, mapped to warp-level primitives without branching outside the parity test.

Compliance. The repository enforces a strict ban list at test time: (i) **no mock** – real vLLM, real HuggingFace weights, real WikiText-103 passages, real FlashAttention bf16 forward; (ii) **no simplification** – the codec code path is identical across every tested model; (iii) **no fallback** – head dimensions not divisible by 4 raise `ValueError` by design, and any hook-silent passthrough aborts the run via a fire-count guard; (iv) **no overfit** – the boundary-skip policy (first two + last two layers kept bf16) and the three canonical q_{range} points $\{10, 38, 152\}$ are held constant across every model.

4 Experimental methodology

4.1 Models and environment

We measure on four open-source decoder-only transformer models spanning four architectural families (Table 1), chosen because each exposes a different challenge: Qwen3-4B has per-head QK-norm; DeepSeek-R1-Distill inherits Qwen2 attention with GQA; Gemma-4-E4B has head_dim = 256 (double the others); GLM-4-9B-Chat has adjacent-pairs RoPE plus QK-norm and a reversed attention-module signature.

Table 1: Benchmark corpus.

Model	Params	L	n_{kv}·hd	Attention family
Qwen/Qwen3-4B	4 B	36	$8 \cdot 128$	GQA + per-head QK-norm
deepseek-ai/DeepSeek-R1-Distill-Qwen-1.5B	1.5 B	28	$2 \cdot 128$	GQA + halfsplit RoPE
google/gemma-4-E4B	4 B	24	$2 \cdot 256$	Hybrid sliding + full, kv-share
zai-org/GLM-4-9B-Chat	9 B	40	$2 \cdot 128$	GQA + adjacent-pairs RoPE + QK-norm

All runs execute on NVIDIA H200 (80 GiB, CUDA 13.0) via vast.ai, with vLLM `0.19.2rc1.dev100+gf9466` and transformers 5.5.2. Weights are loaded in bf16; FlashAttention is enabled.

4.2 Snapshot-mode capture and replace

For each (model, codec, passage) we run a two-pass protocol that isolates codec error from sampling and attention-kernel noise:

1. **Pass 1 (capture).** A real vLLM prefill at ctx $= 2048$ over the passage. A per-architecture attention-module monkey-patch (`snapshot_hook.py`) captures the post-QK-norm pre-RoPE K and V in strict-GPU fp32 (`assert K.is_cuda`; no CPU round-trip).

2. **Offline codec.** Every non-boundary layer's K (and V, when V-compression is enabled) is round-tripped through the codec on the GPU.

3. **Pass 2 (replace).** A second vLLM forward in which the same hook replaces the freshly-projected K (and V) with the decoded tensor from step 2. Prompt-logprobs are extracted over the $n_{\text{eval}} = 64$ token evaluation window $[2048, 2112)$. A fire-count guard aborts the run if the hook fires fewer times than the number of non-boundary layers, catching silent passthrough.

Why snapshot-mode. This mirrors the two-pass `DynamicCache` pattern used in HuggingFace teacher-forcing [49]: each layer's cache entries depend only on the codec applied to the clean Pass-1 capture, not on an error that compounds across layers during a single forward. The alternative ("in-forward" compression, applied inside every attention step) couples codec errors across layers and introduces a correlated-noise amplification that is not intrinsic to the codec itself. Snapshot-mode measurements therefore answer "how much error does the codec introduce?" rather than "how much error does the codec plus the harness introduce?".

4.3 Boundary-layer protection

We keep the first two and last two transformer layers at bf16 ("boundary layers") for every model. These layers have outsized influence on token-level perplexity because they sit at the embedding and pre-softmax boundaries; leaving them uncompressed at negligible cost ($L_{\text{bdry}}/L \in \{10\%, 14\%, 17\%, 10\%\}$ across the four models) meaningfully reduces $|\Delta\text{ppl}|$ while losing $< 15\%$ of the headline compression ratio. The boundary choice is the same constant for all four models (not per-model tuned) to avoid any appearance of overfitting.

4.4 Quality metrics

Three metrics are reported for every (model, codec, q) tuple:

- ***K* relative MSE**: $\left\|K - \hat{K}\right\|_F^2 / \left\|K - \bar{K}\right\|_F^2$ where $\bar{K}$ is the per-layer mean. Sample-size-independent: a closed-form per-vector error.

- **|Δppl|**: absolute relative deviation of the codec pass's perplexity from the bf16 reference, $|\mathrm{ppl}_{\mathrm{alt}} - \mathrm{ppl}_{\mathrm{ref}}|/\mathrm{ppl}_{\mathrm{ref}}$, averaged over 4 WikiText-103 passages $\times n_{\mathrm{eval}} = 64$ eval tokens.
- **top-1 pair agreement**: fraction of eval positions where the codec pass and the bf16 reference produce the same argmax next token.

We use |Δppl| rather than signed Δppl because structured-quantisation codecs exhibit a "de-biasing" phenomenon [30] whereby Hadamard-rotated scalar quantisation slightly lowers ppl below the bf16 reference on well-regularised models. Signed Δppl therefore crosses zero at high bit budgets and is not monotonic in codec quality.

5 Main results

We test four claims in sequence: (i) that the D_4 shaping gain translates to a monotone Pareto improvement across bit budgets (§5.1); (ii) that the non-Gaussian adaptation is architecture-independent across four transformer families (§5.2); (iii) that the measured K-MSE ratio reproduces the theoretical 0.919 to within $\sim 1\%$ (§5.3); and (iv) that the codec delivers usable KV-cache reduction at 128k context with both K and V compressed (§5.4).

5.1 Compression-rate Pareto on Qwen3-4B

Table 2 gives the full Pareto sweep at seven bit levels, matched between TurboQuant $b \in \{2, \ldots, 8\}$ and KakeyaLattice $q_{\mathrm{range}} \in \{2, 5, 10, 19, 38, 76, 152\}$, on four WikiText-103 passages at ctx $=2048$, $n_{\mathrm{eval}} = 64$, strict-GPU.

Table 2: Full Pareto: KakeyaLattice vs. TurboQuant $k8v4$ on Qwen3-4B, H200, vLLM 0.19.2, strict-GPU capture + replace. Bold rows are the three canonical operating points. CR = compression ratio vs. bf16.

bits	CR	Config	Δppl	|Δppl|	top-1	K rel-MSE
288	7.11×	TQ $b = 2$	+150.5%	150.5%	66.4%	5.6×10^{-1}
320	**6.40×**	**v1.4 Q=2**	+30.3%	30.3%	**83.6%**	$\mathbf{1.9 \times 10^{-1}}$
416	4.92×	TQ $b = 3$	+2.22%	5.91%	90.6%	6.7×10^{-2}
448	4.57×	v1.4 Q=5	+2.83%	2.83%	94.5%	3.1×10^{-2}
544	3.76×	TQ $b = 4$	+3.51%	4.22%	97.7%	1.2×10^{-2}
576	**3.56×**	**v1.4 Q=10**	+1.01%	1.86%	97.3%	$\mathbf{7.8 \times 10^{-3}}$
672	3.05×	TQ $b = 5$	+1.24%	1.24%	98.4%	2.7×10^{-3}
704	2.91×	v1.4 Q=19	−0.52%	1.33%	97.7%	2.2×10^{-3}
800	2.56×	TQ $b = 6$	−0.37%	0.96%	98.4%	6.2×10^{-4}
832	**2.46×**	**v1.4 Q=38**	+0.41%	0.57%	**99.6%**	$\mathbf{5.4 \times 10^{-4}}$
928	2.21×	TQ $b = 7$	+0.51%	0.51%	100.00%	1.5×10^{-4}
960	2.13×	v1.4 Q=76	−0.50%	0.67%	100.00%	1.4×10^{-4}
1056	1.94×	TQ $b = 8$	−0.51%	0.66%	98.83%	3.7×10^{-5}
1088	**1.88×**	**v1.4 Q=152**	−0.20%	0.37%	**99.6%**	$\mathbf{3.4 \times 10^{-5}}$

Observations.

1. **K-MSE: the lattice dominates at every bit level** (all seven bit points, ratio 0.35×–0.91×). The ratio grows monotonically with bit budget, consistent with the theoretical prediction that shaping gain is a high-rate asymptote.
2. **Advantage grows with compression aggression.** At 1.88× compression the K-MSE ratio is 0.91×; at 6.4× compression the ratio is 0.35× (65% better). The D_4 shaping gain

matters most in the low-bit regime where scalar quantisation suffers most from boundary-effect truncation.

3. **Catastrophe avoidance at extreme compression.** At the aggressive $q_{\text{range}} = 2$ point (320 bits, 6.4× CR), $|\Delta\text{ppl}|$ is 30.3% with top-1 83.6%, i.e. degraded but recoverable. The lattice structure prevents per-coordinate tail truncation from compounding into total signal loss as the bit budget narrows.

4. **Equal-quality bit savings.** At target K rel-MSE $\approx 5 \times 10^{-4}$, $q_{\text{range}} = 38$ reaches it at 832 bits; interpolating the scalar-quantiser curve gives ∼928 bits for the same K rel-MSE, a ∼10% bit saving at equal downstream quality.

5.2 Multi-model head-to-head

Table 3 gives the 12 head-to-head ratios across four models at the three canonical operating points.

Table 3: Twelve head-to-head pairs. K-MSE ratio < 1 means KakeyaLattice beats the scalar-quantiser baseline on reconstruction; $|\Delta\text{ppl}|$ ratio < 1 means KakeyaLattice is closer to bf16. "top-1 Δpp" is KakeyaLattice top-1 minus baseline top-1. Data: `reports/v1_3_ppl/snapshot_mode_qwen3/multimodel/`.

Model	TQ b	v1.4 Q	CR	**K-MSE ratio**	**$\lvert\Delta\text{ppl}\rvert$ ratio**	**top-1 Δpp**
Qwen3-4B	4	10	3.56×	**0.639**	**0.385**	−0.78
	6	38	2.46×	**0.868**	**0.308**	+0.78
	8	152	1.88×	**0.911**	**0.912**	+0.39
DeepSeek-R1-Distill-1.5B	4	10	3.56×	**0.639**	**0.187**	+3.91
	6	38	2.46×	**0.868**	2.586	−0.78
	8	152	1.88×	**0.911**	3.080	+0.39
Gemma-4-E4B (hd =256)	4	10	3.66×	**0.638**	**0.376**	−0.39
	6	38	2.51×	**0.866**	**0.819**	+0.00
	8	152	1.91×	**0.909**	1.264	−0.39
GLM-4-9B-Chat	4	10	3.56×	**0.639**	**0.551**	+0.78
	6	38	2.46×	**0.868**	**0.637**	−0.39
	8	152	1.88×	**0.911**	**0.490**	−0.39

Metric	KakeyaLattice wins	tie	baseline wins
K-MSE ratio (< 1.0)	**12/12**	0	0
$\lvert\Delta\text{ppl}\rvert$ ratio (< 1.0)	**9/12**	0	3
top-1 pair (≥ baseline)	6/12	2	4

Consolidated verdict. K-MSE dominance is universal: on every model, at every bit level, KakeyaLattice's reconstructed K is closer to the bf16 K than the scalar-quantiser baseline's by 10–36%, mirroring the theoretical ratio $G(D_4)/G(\mathbb{Z}^4) \approx 0.919$ (with the expected monotonic drift to 0.64 at the aggressive $q_{\text{range}} = 10$ point, where lattice shaping matters more).

$|\Delta\text{ppl}|$ dominance at the aggressive point. At $q_{\text{range}} = 10$ all four models beat the baseline by 1.8–5.3×. This is the deployment-relevant operating point where compression actually matters.

$|\Delta\text{ppl}|$ ties at near-lossless. At $q_{\text{range}} = 152$ on DeepSeek-1.5B and Gemma-4-E4B, KakeyaLattice loses on $|\Delta\text{ppl}|$ ratio (values 1.26–3.08). Both absolute numbers are below 1% (baseline 0.39%, ours ∼1%), so the ratio is sensitive to FlashAttention bf16 reproducibility

noise at the four-passage sample size. The K-MSE ratio (0.91×) still shows the K is more faithful to bf16; the $|\Delta\text{ppl}|$ difference is below the bf16 floor on these runs.

5.3 Theory–experiment reconciliation

Table 4: Theory-to-experiment match for the K-MSE shaping-gain ratio. Predicted from Proposition 2.2; measured in three independent environments.

Quantity	Predicted	Measured
K-MSE ratio at matched 1056 bits (Qwen3-4B)	0.919	**0.913** (offline 100k samples)
		0.911 (live vLLM, 4 passages)
		0.911 (multi-model mean, Tab. 3)
Shaping gain (dB)	+0.367 dB	+0.381 dB (Qwen3-4B offline)
$\cos(K, \hat{K})$	$\to 1.0$	1.0000
Bit cost (exact)	1056	1056 (reported rounded to 1088)

All three independent measurements agree with the theoretical prediction to within $\sim 1\%$. We take this as the cleanest theory-to-experiment match the project has produced and as direct empirical validation of Proposition 2.2 at deployment head dimensions.

5.4 128k KV-cache storage

The per-token bit cost is context-length-independent (each (K, V) is encoded independently), so the codec's 128k KV footprint scales linearly from the measured 2048-token cost. Table 5 reports the three canonical operating points with K *and* V both compressed (boundary layers bf16).

Table 5: 128k KV cache storage: baseline bf16 vs. KakeyaLattice (both K and V compressed, boundary layers $\{0, 1, L-2, L-1\}$ kept bf16). Per-user GiB at ctx = 128,000. Source: `reports/v1_4_release/kv_128k_report/V14_KV_128K_REPORT.md`.

	bf16 128k KV		KakeyaLattice 128k KV			
	GiB	ratio	Q=10	Q=38	Q=152	$\lvert\Delta\text{ppl}\rvert$ best row
Qwen3-4B	18.0	1.0×	2.77×	2.12×	1.71×	0.48% @ Q=152
DeepSeek-R1-Distill-1.5B	3.5	1.0×	2.60×	2.04×	1.67×	0.28% @ Q=152
Gemma-4-E4B (hd =256)	6.0	1.0×	2.53×	2.01×	1.66×	0.15% @ Q=38
GLM-4-9B-Chat	5.0	1.0×	2.83×	2.15×	1.73×	0.90% @ Q=152

Quality at the balanced Q=38 point. $|\Delta\text{ppl}| < 1\%$ for every model except GLM-4-9B-Chat (2.51%); top-1 $\geq 95\%$ universally. This is the deployment-recommended operating point: $\sim$2.1× real KV reduction on a 9B-class model, essentially lossless on the first three families.

Quality at the near-lossless Q=152 point. $|\Delta\text{ppl}| < 1\%$ on every model; top-1 $\geq 98\%$ on every model. Recommended when quality is paramount and $\sim$1.7× is acceptable.

Aggressive Q=10. GLM-4-9B-Chat is past its edge at 6.5% Δppl; safe on Gemma-4-E4B (0.33%); noticeable on Qwen3-4B and DeepSeek (1.45%, 2.22%).

5.5 Codec wall-clock

Table 6: Encode time ratio v1.4 / TQ $k8v4$, per 2112-token passage on H200. Values < 1.0 mean KakeyaLattice is faster.

Operating point	Qwen3-4B	DeepSeek-1.5B	Gemma-4-E4B	GLM-4-9B	Mean
Q=10 (aggressive)	0.80	0.77	1.18	0.81	**0.89**
Q=38 (balanced)	1.53	1.62	1.45	1.69	1.57
Q=152 (near-lossless)	1.48	1.63	1.47	1.58	1.54

At the aggressive point the lattice codec is $\sim 10\%$ faster than the scalar-quantiser baseline on three of four models. At Q= 38 and Q= 152 it is $\sim 55\%$ slower; this is the cost of the D_4 closest-point search plus per-vector qmax cycle versus the baseline's simpler rotation + quantise. Absolute codec wall time is 0.02–0.04 s per 2112-token passage — negligible against the ~ 10 s of bf16 prefill at this ctx length. Speed is not a deployment blocker.

6 Caveats and known limits

1. **Four-passage sample size.** $|\Delta\text{ppl}|$ differences below $\sim 1\%$ absolute are inside the FlashAttention bf16 reproducibility floor at this sample size. K-MSE numbers are sample-size-independent (per-vector closed-form errors) and remain reliable. Publication-grade $|\Delta\text{ppl}|$ tightening would require ≥ 32 passages.

2. **Snapshot-mode vs. in-forward.** All numbers in this paper are snapshot-mode (§4.2), which isolates the codec's intrinsic error. In-forward integration (codec inside every attention step) adds correlated per-layer noise that is not intrinsic to the codec; measuring that accurately needs a paged-attention-native v1.4 kernel, which is out of scope for this paper.

3. **Head_dim $\nmid$ 4.** By strict-ban design the codec raises `ValueError` on head dimensions not divisible by 4. Models with $d_h = 96$ or $d_h = 176$ would need a different lattice (e.g. E_8 blocks for d_h divisible by 8).

4. **Kv-sharing layers.** Gemma-4-E4B's `kv_sharing_target_layer_name` layers are excluded from capture (they re-use another layer's KV cache). This matches the model's native semantics at zero PPL cost.

5. **Kakeya lower bound is conditional at $D \geq 4$.** Proposition 2.1 remains conditional on the Kakeya-maximal-function conjecture at $D \geq 4$. At $D = 3$ the lower bound is unconditional via Wang–Zahl [21]; at deployment head dimensions it is a conditional framing, not a proof of Pareto-optimality. The D_4 *shaping bound* in Proposition 2.2 is unconditional, but it applies to the 4-dimensional block structure of the codec rather than to the full head dimension.

7 Comparison with TurboQuant

KakeyaLattice is designed to live inside an existing scalar-quantisation framework rather than to replace one. This section gives the head-to-head positioning with TurboQuant [34, 35], the strongest post-hoc KV codec currently deployed in a production inference stack.

7.1 What TurboQuant is

TurboQuant is a KV-cache compression family merged into vLLM [35] in April 2026 and subsequently hardened by a series of follow-up PRs [36, 37, 38, 39]. The canonical preset, $k8v4$,

compresses K with an FP8 E4M3 scalar quantiser after a Sylvester–Hadamard rotation and unit-norm factorisation, and V with a 4-bit uniform quantiser. Slots are rounded to the next even number rather than to the next power of two, which at $D = 128$ gives 1056 bits/token/head for K ($8 \cdot 128 + 2 \cdot \text{fp16} = 1056$).

7.2 The four engineering levers in common

The engineering stack of TurboQuant and KakeyaLattice overlaps in four levers: unit-norm factorisation, Sylvester–Hadamard rotation, per-vector adaptive $q_{\max}$, and a joint (per-vector shared) scale. The lattice codec uses these four levers precisely because scalar-quantisation literature has shown they dominate the MSE accounting for any Gaussian-like residual source; adopting them matches the codec's input distribution to the isotropic-Gaussian assumption of Proposition 2.2. Without these levers the predicted 0.919 K-MSE ratio does not manifest.

7.3 The one lever that differs

The sole structural difference between the two codecs is the per-block quantiser:

	TurboQuant $k8v4$	KakeyaLattice
per-block quantiser	int·per-coord ($\mathbb{Z}^4$)	D_4 closest-point
block size	1 coordinate	4 coordinates
second moment	1/12	0.0766
shaping gain	0 dB (reference)	0.367 dB

Everything else in the stack — Hadamard matrix, per-vector $q_{\max}$, unit-norm factorisation, bit alignment, boundary layer policy, fp16 overhead scalars — is identical by construction. Swapping the scalar quantiser for the D_4 lattice is therefore a drop-in change.

7.4 Positioning: drop-in improvement, not replacement

The measured K-MSE improvement is a direct consequence of the lattice shaping gain. It is unconditional (see Proposition 2.2) and architecture-independent (see Table 3: the ratio is within $\sim 1\%$ of 0.919 on every tested model). At matched bits, the other lever contributions cancel exactly because they are identical between the two codecs; the residual is the shaping gain.

We therefore position KakeyaLattice as an $\approx 8\%$ *improvement to the upstream TurboQuant kernel*, not as a standalone KV codec. Every TurboQuant deployment becomes $\sim 8\%$ more faithful on K at zero additional engineering cost, plus the aggressive-compression 1.8–5.3 $\times$ $|\Delta\text{ppl}|$ improvement at the $q_{\text{range}} = 10$ operating point.

7.5 Proposed upstream PR scope

- A Triton kernel for the Conway–Sloane D_4 closest-point algorithm ($\sim$10 lines of element-wise ops).
- Slot-format alignment: 32×32 bits of lattice plus two fp16 overhead scalars per vector, bit-compatible with the existing $k8v4$ slot layout.
- Register as a new `--kv-cache-dtype` preset, e.g. `turboquant_d4v4`, alongside `turboquant_k8v4`.

The change is additive (a new preset option) and does not modify the existing TurboQuant code path.

7.6 Compression vs. per-bit fidelity

At each matched bit point, KakeyaLattice pays $\sim$ 3–11% more bits (from the $O(q_{\text{range}}^3)$ boundary correction encoded as a clamp step) for 8–36% better K-MSE. The *Pareto curve* is strictly better (Table 2); the *raw compression ratio at a fixed operating point* is slightly worse. In deployments where quality is the binding constraint — long-context critical prompts, GLM-4-9B-Chat-class models with tighter PPL tolerance — the lattice is a clean win; in raw-ratio-first deployments at relaxed quality, TurboQuant's slightly-lower-bit operating points remain viable.

7.7 End-to-end vLLM throughput (not measured here)

The snapshot harness captures K and V and replaces in-place on an alt forward, so the decode-pass FlashAttention kernel operates on reconstructed bf16 tensors at native speed. Only the codec itself costs wall time (Table 6: 0.02–0.04 s per 2112-token passage at the balanced point). A production throughput comparison requires a paged-cache backend that embeds the v1.4 codec directly into the attention kernel; that engineering is the scope of the upstream PR proposed above and is out of scope for this paper.

8 Related work

8.1 Kakeya problem and harmonic analysis

The Kakeya problem was posed by Kakeya in 1917 [1] as a question about the minimum measure of a region in which a unit segment can be rotated continuously. Besicovitch gave the first nontrivial Kakeya-type constructions in 1919 [2] and 1928 [3], showing sets of arbitrarily small planar measure containing a line segment in every direction. Fefferman's resolution of the ball multiplier problem [4] was the first application connecting Kakeya-type sets to oscillatory-integral bounds; Córdoba [5] developed this further. Bourgain opened the arithmetic-combinatorial era with a nontrivial high-D bound [6]; Wolff introduced the geometric hairbrush argument and pushed the $\mathbb{R}^3$ bound forward [7]; Katz, Łaba, and Tao sharpened the bound for Besicovitch sets in $\mathbb{R}^3$ [8]; Katz and Tao investigated the discretized setting [9]. Tao surveyed the landscape through approximately 2007 [10]. Dvir resolved the finite-field analogue with a one-page polynomial-method proof [11]; Dvir, Kopparty, Saraf, and Sudan extended the technique to list-decoding and extractors [12]. Bennett, Carbery, and Tao posed the multilinear Kakeya conjecture [13], Guth resolved its endpoint case [14], and Guth–Katz [15] resolved the distinct-distances problem in the plane using closely related polynomial-partitioning tools. Hickman, Rogers, and Zhang pushed high-D bounds to near Kakeya-conjectural levels [18]. Zahl investigated restricted Kakeya-type problems [19, 20]. In 2025 Wang and Zahl [21] resolved the three-dimensional Euclidean Kakeya conjecture.

The paper's positioning relies on three specific links in this literature: Guth's multilinear Kakeya endpoint [14], the Brascamp–Lieb equivalence of Bennett–Carbery–Christ–Tao [16], and the direct Brascamp–Lieb-based matrix-concentration proof by Carbery and Valdimarsson [17]. Tropp's user-friendly matrix tail bounds [22] are the concrete form of matrix concentration we rely on.

8.2 Lattice source coding

Zamir and Feder's analysis of lattice quantisation [24] established the basic rate–distortion framework used in this paper. The D_4 root lattice's second-moment optimality in $\mathbb{R}^4$ is a consequence of Conway–Sloane's closest-point analysis [25] and their general treatment [26]. Eyuboglu–Forney [27] formalised the memoryless-source high-rate theory of lattice-bounded codebooks; Forney's trellis-coded modulation work [28] and Erez–Zamir's capacity-achieving lattice construction [29] are closely related in spirit. Chee, Cai, Kuleshov, and De Sa [30] applied

lattice codebooks to weight quantisation for LLMs via QuIP#; our codec applies the same structural idea at the KV-cache level with the specific combination of unit-norm factorisation, Hadamard rotation, per-vector adaptive scale, and D_4 closest-point.

8.3 Randomised numerical linear algebra

The Halko–Martinsson–Tropp randomised SVD [23] closes the Brascamp–Lieb to matrix-concentration arc in numerical form. It does not appear in the v1.4 codec (the RSVD-based v1.3 codec used it extensively [33]), but it is cited because the proof of Proposition 2.2 rests on matrix-concentration bounds that are derivatives of Tropp's matrix Chernoff inequality [22], whose sharpest form goes through Carbery–Valdimarsson [17].

8.4 KV-cache compression

Token eviction. H_2O [42] and Scissorhands [43] propose attention-score-driven token eviction at test time; StreamingLLM [44] preserves a small prefix of attention-sink tokens exactly and streams the rest. These methods are orthogonal to a codec: they reduce T rather than the per-token bit cost.

Per-channel quantisation. KIVI [40] is a per-channel asymmetric 2-bit quantiser on K combined with per-token on V. Its rotation-free structure sits outside the Hadamard-rotation stack of TurboQuant and KakeyaLattice.

Hadamard-based codecs. *Cache Me If You Must* [41] applies Hadamard rotation combined with adaptive per-token bit-width allocation. TurboQuant [34, 35] provides the production-deployed Hadamard + Lloyd-Max + norm-correction stack we benchmark against (§7); follow-up vLLM PRs include [36, 37, 38] with NIAH long-context validation in [39].

Trained low-rank methods. Dynamic Memory Compression [45] and LESS [46] learn low-rank projections during training; they are orthogonal to the post-hoc codec line studied here.

Prior work in this line. [33] is the prior RSVD + Lloyd-Max codec in our repository. Its rank cap $r = D/2$ relied on a conjectural Kakeya-maximal-function lower bound in dimension $D \geq 4$; the present codec replaces the conjectural bound with the unconditional D_4 shaping bound of Proposition 2.2.

8.5 LLM inference infrastructure

vLLM [48] is the reference serving stack used throughout this paper; its prefix-caching and block-manager abstractions make snapshot-mode measurement possible. The transformers [49] library supplies the weights and tokenisers. FlashAttention [50] is the default attention kernel in all forward passes reported here.

9 Conclusion

We have presented KakeyaLattice v1.4, a Zamir–Feder D_4 nested-lattice codec whose shaping gain over scalar quantisation is bounded unconditionally by $G(\mathbb{Z}^4)/G(D_4) \approx 0.919$ via lattice density. Placing the construction inside the Kakeya–Brascamp–Lieb–Tropp dependency chain

(§2) grounds the design in a literature-traceable framework: multilinear Kakeya [14] → Brascamp–Lieb [16] → matrix concentration [22] → lattice second-moment optimality [26], with the three-dimensional Kakeya case [21] at the chain's deep end and our codec at its shallow end. The rate–distortion boundary argument of Proposition 2.2 is unconditional at the codec's 4-dimensional block structure, in contrast to the conditional lower bound of our prior work [33].

The measured results are consistent across four transformer families. On twelve head-to-head pairs at three matched bit points the codec wins universally on K-MSE (12/12), at nine of twelve pairs on $|\Delta\text{ppl}|$, and at all four models at the aggressive $\sim 3.6\times$ compression point. The theoretical shaping-gain prediction of 0.919 is recovered to within $\sim 1\%$ on every tested model in three independent measurement environments. At 128k context the codec reduces the combined KV cache by 2.5×–2.8× at a balanced operating point and 1.66×–1.73× at a near-lossless operating point.

Our positioning (§7) is as an unconditional $\approx 8\%$ improvement to an existing production-deployed KV kernel, exposed as an additive upstream PR. The change is a single per-block quantiser substitution — D_4 closest-point in place of int·per-coord uniform — inside an otherwise unchanged engineering stack, and the measured gain is architecture-independent across four open-source model families.

Reproducibility. All code, unit tests, per-model JSON measurements, the four per-architecture attention hooks, and the multi-model head-to-head harness are released under Apache-2.0 at tag `v1.4`. The canonical codec class is `kakeyaturbo_py.V14KakeyaZamirLatticeGPU`; the multi-model report is at `reports/v1_3_ppl/snapshot_mode_qwen3/multimodel/FINDINGS_MULTIMODEL.md`; and the 128k storage report is at `reports/v1_4_release/kv_128k_report/V14_KV_128K_REPORT.md`.

A note on the distortion metric. Throughout this paper we report reconstruction MSE and its relative normalisation rather than inner-product distortion. For any unit-norm query q, $\left|q^\top(k-\hat{k})\right| \le \left\|k-\hat{k}\right\|$, so an MSE bound is a conservative upper bound on the attention-logit perturbation that matters for downstream quality. The measured $|\Delta\text{ppl}|$ and top-1 agreement corroborate that the MSE proxy is tight in practice.

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A Reproducibility manifest

The full multi-model benchmark is reproducible via:

```
cd /workspace/LLM-KV--Cache-compress
export VLLM_ENABLE_V1_MULTIPROCESSING=0 KAKEYA_SNAPSHOT_QWEN3=1
python benchmarks/multimodel_v14_vs_tq.py \
    --model-path Qwen/Qwen3-4B --model-name qwen3_4b \
    --ctx-len 2048 --n-eval 64 --n-passages 4 \
    --out-dir reports/v1_3_ppl/snapshot_mode_qwen3/multimodel
```

Replace `--model-path` / `--model-name` for the other three models; add `--trust-remote-code` for GLM-4-9B-Chat. The 128k storage report uses `benchmarks/multimodel_v14_kv_128k_report.py` with the same capture/replace protocol and both K and V compressed.

B Canonical naming

- **Spoken / written name**: "v1.4 kakeya zamir lattice GPU"
- **Short form with parameter**: e.g. "v1.4 Q=152"
- **Class name**: `V14KakeyaZamirLatticeGPU`
- **Module**: `kakeyaturbo_py.v1_4_kakeya_zamir_lattice_gpu`

C Canonical operating points

q_{range}	bits/tok/head	CR (bf16)	Role
152	1088	1.88×	Near-lossless
76	960	2.13×	
38	832	2.46×	**Deployment recommended balanced**
19	704	2.91×	
10	576	3.56×	Aggressive compression
5	448	4.57×	
2	320	6.40×	Extreme compression